ALEX FERGUSON

SENIOR SOFTWARE ENGINEER

Backend · AI · Agentic Engineering

South Wales, UK · 07841 384088
hello@awlf.me · www.awlf.me

Senior software engineer with a backend focus, working across Python, TypeScript and AWS. Currently leading Veygo's Agentic Guild and building production workflows that use LLMs to plan, implement, review and operate software.

EXPERIENCE

Veygo

Senior Software Engineer
Jul 2025 – Present · Cardiff / Hybrid

Software Engineer
Aug 2022 – Jul 2025

- Lead the Agentic Guild, setting practical standards for agentic engineering, helping engineers adopt AI confidently, and shaping how the company uses AI in production.
- Co-created Veygo's agentic SDLC, taking work from ticket through exploration, planning, implementation, testing, hardening and PR, with human approval points built into the workflow.
- Designed and built a Bot Platform for production Slack workflows using Python, ECS, SQS, Lambda, API Gateway and Bedrock. Agents use pre-scoped tools rather than unrestricted API access.
- Built production agent workflows for New Relic incident triage and Customer Care referrals, handling hundreds of alerts and up to 10 referrals per week.
- Led a payments rewrite from an unreliable Django implementation to a Stripe facade API using DynamoDB and Temporal. Reduced errors from hundreds per week to single digits and brought a product back on sale.
- Own end-to-end delivery from architecture and implementation through infrastructure, CI/CD, production monitoring and incident resolution.
- Mentor engineers, support onboarding and improve engineering processes across squads.

SELECTED IMPACT

PAYMENTS **100s → <10 errors per week**	AGENTIC SDLC **Ticket → PR workflow**
BOT PLATFORM **5 production workflows**	INCIDENT BOT **Hundreds of New Relic alerts**

HOW I WORK

I like building things that are slightly harder than they first appear. I'm strongest when I can own a problem end to end, make the architecture simpler, and leave the system easier to operate than I found it.

I'm particularly interested in the point where traditional software engineering meets agentic systems: what should an agent be allowed to do, how do we measure whether it is safe to let it do more, and where does human review actually add value?

TECHNICAL

AI & Agentic
Claude · Amazon Bedrock · agent orchestration · multi-step workflows · guardrails · agentic SDLC

Backend
Python · Django · TypeScript · REST APIs · SQL · Temporal

AWS
ECS · Lambda · SQS · API Gateway · DynamoDB · RDS · S3 · IAM

Platform
Docker · CI/CD · production monitoring · incident response

Testing
Unit · integration · contract · mutation testing · automated hardening · security tooling

CORE STRENGTHS

Architecture: Backend systems, APIs, queues and workflow orchestration.

Delivery: From idea and architecture through implementation, CI/CD, infrastructure and production.

Agentic engineering: Practical LLM workflows, scoped tools, guardrails, verification and human-in-the-loop design.

Leadership: Technical direction, mentoring, pairing and making complex decisions easier for teams.

EDUCATION

University of East Anglia
BSc Computing Science with Year in Industry
First Class (4.5 GPA) · 2017–2021
Modules: AI, machine learning, secure software, Python, Java, C/C++, data structures, networks.
Project: Text weather forecast generation using ML in Python.

A closer look at the problems behind the headline work. I tend to work across architecture, implementation and operations rather than stopping at the code change.

Agentic SDLC	Designed the architecture and workflow for an agentic delivery process. Claude-generated implementation is checked through proposal approval, implementation tests, hardening agents, agent review, human PR review and existing CI/security controls. The workflow covers ticket → deploy, with CircleCI taking over after merge.
Veygo Bot Platform	Built the platform used to run multiple Slack-based agents. Lambda/API Gateway formats and queues work, SQS feeds ECS workers, and Amazon Bedrock provides the model. Agents use pre-scoped tools with predetermined permissions. Current focus includes improving memory, retries, observability and evaluation.
Production Incident Automation	Built a New Relic workflow that listens to alerts, investigates likely causes using production signals, recent PRs and code search, and can either propose a code change or tune alert sensitivity. Code changes require a PR and human review; alert tuning has an explicit human sign-off point.
Customer Care Automation	Built a workflow that reads customer-care referrals, searches an internal knowledgebase, answers straightforward questions or routes work to the right team, and follows up on unresolved referrals so the knowledgebase can improve over time.
Payments Rewrite	Replaced a fragile payments implementation embedded in the Django monolith and tied to an unsupported Stripe API version. The replacement is a deliberately small Stripe facade with DynamoDB and Temporal orchestration, giving product teams a consistent payment API. The result reduced errors from hundreds per week to single digits and restored a product that had been taken off sale.

EARLIER EXPERIENCE

Aforza
Associate Software Engineer
Jul 2021 – Aug 2022 · Cardiff
Node.js · TypeScript · NestJS · Docker · Kubernetes · REST APIs. Built backend services and owned features through delivery.

Met Office
Scientific Software Engineer (Industrial Placement)
Jul 2019 – Jul 2020 · Exeter
Python 3 · SciPy · weather-model data · testing. Developed post-processing software and automated test suites.

Robot Mascot
Junior Developer
Nov 2017 – Feb 2019 · Cardiff
PHP · JavaScript · HTML · Linux. Built bespoke websites and maintained hosted applications.

ENGINEERING APPROACH

Architecture
Start with the simplest design that gives the system a clean boundary. Use queues, workflows and services where they solve a real problem, not because they are fashionable.

Agentic Systems
Treat agents as software with permissions and failure modes. Prefer scoped tools, explicit approval points and strong verification over broad access.

Delivery
Get the full path working: implementation, tests, CI/CD, infrastructure, monitoring and production support. A feature isn't finished when the PR is merged.